

Experiments 4A-C: Synthesis and Component Analysis of an Iron (III) Oxalate Complex

My Name

TA, Section B##

Work Performed on 10/23, 10/30, & 11/4, 200#

Report due Tuesday, November ##, 200#

1. Abstract

This experiment initially involved the synthesis of an iron (III) oxalate complex with the general formula $K_w[Fe_x(C_2O_4)_y] \cdot zH_2O$. The variables x , y , and z were determined through the duration of the entire experiment. From 1.2000g of $Fe(NH_4)_2(SO_4)_2$ were synthesized 1.1###g of $K_3[Fe^{III}(C_2O_4)_3] \cdot 3H_2O$, owing for a percentage yield of 74.###%. A percentage yield of 11#.##% was also calculated, had the final product been $K[Fe^{III}(C_2O_4)_2] \cdot 2H_2O$. This value was rejected because experimental errors are never to exceed theoretical values, which are taken on assumption that the experiment is carried out in entirely ideal environmental settings, which are impossible to attain entirely. In 4B, redox titrations were used to calculate the relative composition of oxalate within the iron (III) oxalate salt. The millimoles of oxalate were found to be .64#, .63#, .62#, and .62#, for trials A-D, respectively. The weight percentages for each trial (A-D) were 54.0#%, 53.4#%, 54.1#%, and 53.##%, with an average of $53.## \pm .5###\%$. It was also found that the four trials yielded calculations of 61#, 60#, 61#, and 61# mmol of $C_2O_4^{2-}$ per 100g of complex analyzed. The averaged value, with the inclusion of two standard deviations, reported to be $61# \pm 7.##$ mmol of $C_2O_4^{2-}$ per 100g of complex analyzed. In 4C, Spectrophotometry was used to determine the iron content in the same iron (III) oxalate complex. The average weight percentages of Fe^{3+} for samples 3-5 of the unknowns were found to be 12.##%, 12.7#%, and 14.##%, with an average weight percentage of $13.09 \pm 2.58\%$. The number of millimoles present in 100g of complex were found to be, for unknown samples 3-5: 215.### mmol Fe^{3+} , 227.### mmol Fe^{3+} , and 259.### mmol Fe^{3+} , with an average of $23#.### \pm 4#.##$ mmol Fe^{3+} in every 100g of complex. This data was used, along with the data from 4A and 4C, to conclude that the empirical formula of the complex was $K_3[Fe(C_2O_4)_3] \cdot 2H_2O$, and the experiment had a percent yield of 74.##%.

2. Introduction

The goal of the three experiments is to successfully synthesize an iron (III) oxalate compound and then further analyze it for oxalate (4B) and iron (4C) composition.

A coordination compound is comprised of at least one complex ion, and a complex ion itself consists of two parts: a [transition] metal ion (which has a positive charge, making it an electron-deficient Lewis acid) and one or more ligands (molecules or anions usually of negative charge, rendering them electron-rich Lewis bases, through some neutral molecules such as H_2O and NH_3 work as well). In the union of the electron-deficient metal ion and the electron-rich ligand, the two species are said to “coordinate” to each other. The coordination number of the metal specifies the number of electron pairs which are donated to the metal center. Coordination numbers may vary, with 2, 4, and 6 being the most common.

Ligands themselves may vary in the number of available binding sites. Monodentate ligands are those which donate a single pair of electrons to the metal center. Hence, six monodentate ligands – such as H_2O – bound to a center metal ion would result in a coordination number of six. “Chelate” and “multidentate” are the terms applied to ligands capable of binding at more than one site. Bidentate ligands (such as the oxalate ion $C_2O_4^{2-}$ used in this experiment) bond at two sites on the center metal ion, and tridentate ligands (such as EDTA or ethylenediaminetetraacetato-) bind at three sites. The entire complex may have an overall charge that is positive, negative, or even neutral.

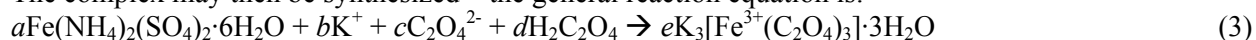
Reduction-oxidation reactions (or redox, for short) are ones in which there is a change in the oxidation numbers of two or more reactant-product pairs. The oxidized component is said to lose electrons (i.e., Fe^{2+} is oxidized to Fe^{3+}), becoming more positively charged, while the reduced component

is said to gain electrons (i.e., Sn^{4+} gains two electrons to be reduced to Sn^{2+}), becoming less positively charged. The redox components occur in pairs, with one species (the reductant) giving up its electron(s) to be taken up by the other species (the oxidant).

4A involves the synthesis of an iron (III) oxalate complex. The iron is first presented in its Fe^{2+} form, so it must first be oxidized to Fe^{3+} before the oxalate ion will bind readily to it. Hydrogen peroxide is the oxidant of choice: $2\text{Fe}^{2+}(\text{aq}) + \text{H}_2\text{O}_2(\text{l}) + 2\text{H}^+(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$, in acidic solution. (1) The oxalate ion is then free to coordinate to the Fe^{3+} ion, forming a complex of $\text{Fe}(\text{C}_2\text{O}_4)$. The oxalate ion is the conjugate base of the weak oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$. In lab, careful attention is paid to the maintenance of the $\text{H}_2\text{C}_2\text{O}_4\text{-C}_2\text{O}_4^{2-}$ equilibrium, as only the oxalate ion is capable of coordinating. This is done through the addition of $\text{H}_2\text{C}_2\text{O}_4$ and the soluble $\text{K}_2\text{C}_2\text{O}_4$ – only the oxalate anion is of concern:

(2)

The complex may then be synthesized – the general reaction equation is:

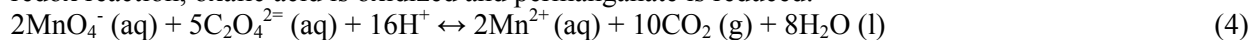


The complex is then precipitated out and rendered into a crystalline form with ice (decreased temperature results in decreased solubility) and ethanol (a polar solvent; the nonpolar target product will have limited solubility in ethanol because of the violation of the like-dissolves-like rule).

The most likely synthesis products are $\text{K}[\text{Fe}^{3+}(\text{C}_2\text{O}_4)_2] \cdot 2\text{H}_2\text{O}$ and $\text{K}_3[\text{Fe}^{3+}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$. The complex with a percent yield (see Expt. 4A calculations for pertinent equation) of $>100\%$ will be rejected, for experimental values will never 100% of the calculated theoretical percentage yield.

Experiment 4B invoked the use of redox titrating to determine the oxalate content ($\text{C}_2\text{O}_4^{2-}$) of the salt. An aqueous solution of the iron oxalate salt is prepared for titration, and the oxalate content is found through analysis by measuring the volume of titrant ($.0100\text{M KMnO}_4$) needed to react completely with the known volume of analyte, or the species to be titrated. Here, it is the aqueous solution of the iron oxalate salt. The stoichiometric (or equivalence) point is the point at which equivalent stoichiometric amounts of the two species have reacted. A secondary indicator is often used to provide a visual clues; the indicator will change color at its endpoint. The defining characteristic of a good indicator is one that has its endpoint as close to the reaction's stoichiometric point as possible. In Experiment 4B, a secondary indicator is not needed, for the intensely-colored permanganate solution serves as its own indicator through a redox reaction. The product Mn^{2+} has a faint pink color, which serves as the visual indicator that the stoichiometric point has been reached.

The permanganate ion is a strong oxidizing agent capable of oxidizing oxalic acid to carbon dioxide and water used to determine the concentration of oxalate ions within the synthesis product. In the redox reaction, oxalic acid is oxidized and permanganate is reduced:



The reaction is best when performed in a strongly acidic environment, which allows the oxalate ions to be present in their fully-protonated form as oxalic acid, $\text{H}_2\text{C}_2\text{O}_4$.

Experiment 4C utilized Spectrophotometry to determine the iron content in the iron (III) oxalate complex. The results were combined with findings from 4A and 4B to determine the empirical formula and percentage yield of the compound synthesized in 4A.

Spectrophotometry is a routine laboratory test that has the added advantage of being able to analyze multiple samples in a short amount of time. The sample to be tested absorbs a certain amount of electromagnetic radiation from a source, and the amount absorbed is related to the concentration of a species in solution—or more specifically, the concentration (c) and the path length (b). the ratio of intensity of light entering the cell (I_0) and leaving the cell (I) are related by Beer's Law. The transmittance (T) represents the fraction of the original radiant energy that passes through, or is transmitted, to the other wise of the cell. These terms are related by the equation:

$$\ln\left(\frac{1}{T}\right) = \ln\left(\frac{I_0}{I}\right) = kbc \quad \text{or} \quad \log\left(\frac{1}{T}\right) = \log\left(\frac{I_0}{I}\right) = \epsilon bc \quad (5)$$

The term I_0/I is renamed absorbance where $A = \epsilon bc$ with $A = \log(1/T)$. The conversion between absorbance and percent transmittance is given by $A = \log\left(\frac{100}{\%T}\right)$. The most efficient way of determining

concentration is to prepare a set of standard solutions of known concentration, record the absorbance readings at a fixed wavelength, and plot the absorbance vs. concentration data. The slope of the line is ϵb . Samples of unknown concentration are run through the same test, under identical conditions, so that absorbance, and eventually concentration, may be determined. The wavelength of 520 nm selected for experiment 4C was because this is the maximum absorbance for the iron (III) ion. Transmittance readings should be limited to within the range of 12.5% to 80% to minimize errors.

The iron complex solution prepared in this experiment is light green, with a complementing color of red-orange (wavelength roughly 600nm). The color intensity is relatively weak, so the hue is intensified by the addition of the bipyridyl ligand, which absorbs ultraviolet radiation and appears colorless. When the ligand coordinates with iron using the lone pair electrons from the nitrogens, the absorption of light shifts into the visible spectrum and the complex appears more intensely colored (to the human eye). In this experiment, the iron (III) is released by the addition of Ca^{2+} , which helps to precipitate out the $\text{C}_2\text{O}_4^{2-}$, leaving behind the iron (III). The solution is centrifuged to remove this precipitate, leaving the iron ion behind to complex with the bipyridyl. Ascorbic acid is used as a reducing agent because it does not absorb in the visible spectrum of light.

The reaction below occurs in the pH 2-8 range, so buffer is added to maintain the pH near 4.7.

3. Experimental

3A. General Information and Materials

All pre-synthesized chemicals were obtained from the York 3150 chemistry stockroom. The iron oxalate coordination complex was prepared in accordance with the instructions provided in the lab manual.¹ All reactions were carried out in the York ##### laboratory classroom under normal atmospheric conditions. The chemicals, listed in order of use, were: $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$, H_2O , 6M H_2SO_4 , 1M $\text{H}_2\text{C}_2\text{O}_4$, 2M $\text{K}_2\text{C}_2\text{O}_4$, 6% H_2O_2 , 95% $\text{C}_2\text{H}_5\text{OH}$, 0.01M KMnO_4 , 1M NaOH , $\text{C}_6\text{H}_8\text{O}_6$, 0.0020 bipyridyl solution, pH 4.7 buffer, 9M H_2SO_4 , 0.5M CaCl_2 . The major laboratory apparatus used, apart from the standard laboratory balances, ovens, and glassware, was the Spectronic 20 spectrophotometer.

3B. Synthesis of the Iron-Oxalate Coordination Complex

1.2000g of the hydrated salt $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$ (3.0600×10^{-3} mol) were obtained and combined with 3 mL of deionized water and 2 drops of 6M H_2SO_4 in a 50 mL beaker. The solution was swirled to dissolve the salt, and heated in a water bath to complete the dissolution. Using a 15 mL graduated cylinder, 6 mL of 1M $\text{H}_2\text{C}_2\text{O}_4$ were measured into the beaker, forming a yellow precipitate. The solution pH was confirmed to be strongly acidic, and 6M H_2SO_4 was added if necessary. A water bath was set up and the solution was swirled for several minutes to digest the precipitate, then the beaker was removed from the heat and allowed to cool to room temperature. The resulting supernatant was then removed using a dropper. The remaining precipitate was washed three times with 3 mL volumes of hot

deionized water, with the waste put into the proper receptacle. 3 mL of 2M $\text{K}_2\text{C}_2\text{O}_4$ were added and the mixture was heated to 40°C. 10 mL of 6% H_2O_2 were then slowly added dropwise, with constant stirring and monitoring of the 40°C solution environment. If there is not a brown solid of $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ at this point, more peroxide was added. The mixture was then heated to boiling, removed from heat, and had added to it 2 mL of 1M $\text{H}_2\text{C}_2\text{O}_4$. The solution was then stirred, and 15 drops more of 1M $\text{H}_2\text{C}_2\text{O}_4$ were added. There should have been a yellow-green solution of dissolved $\text{K}_w[\text{Fe}_x(\text{C}_2\text{O}_4)_y] \cdot z\text{H}_2\text{O}$. If not, 12 drops each of 2M $\text{K}_2\text{C}_2\text{O}_4$ and 1M $\text{H}_2\text{C}_2\text{O}_4$ were added. If still not yellow-green, several drops of 6% H_2O_2 were added, the solution was heated gently, and further assistance was sought. The product was then crystallized by introducing 6 mL of 95% $\text{C}_2\text{H}_5\text{OH}$ (EtOH). If crystals formed in massive amounts, they were redissolved upon warming of the solution. If few crystals formed, 25 drops more of EtOH were added to increase crystallization. The beaker was then stored in foil and stored securely to allow for the completion of crystallization. At least 36 hours of waiting were required in between procedures. The mass of a clean and dried crucible was determined. Using a filtering station setup with a vacuum trap, the crystals were filtered, air-dried for a few minutes, and then washed twice with small volumes of 95% EtOH. The vacuum was continued until the alcohol odor was gone. The crystals were then allowed to air-dry for at least 30 minutes. The mass of the filled crucible was taken, and the mass of the crystals calculated. A minimum of 0.5 g of the neon-green, sand-like crystals were necessary for continuation of the experiment.

3C. Analysis of the Oxalate Ligand Content through Redox Titration

Using weighing paper, four 0.1000 g samples of the synthesized green crystals were obtained and distribute among four 125 mL Erlenmeyer flasks. 30 mL of deionized water and 5 mL of 6M H_2SO_4 were added to each flask, which was then swirled to dissolve the solid. Using a clean dry 100 mL beaker, 90 mL of the standardized KMnO_4 solution was obtained and the exact concentration noted. A buret was obtained and labeled, and thoroughly cleansed it and checked it for any defects. It was then rinsed twice with 5 mL volumes of deionized water and 2 mL volumes of the permanganate solution, then filled to approximately the 0.00 mL mark. Reading the top of the meniscus, the initial volume was recorded. One of the Erlenmeyer flasks (with the prepared solution) was heated to 60°C and the titration with KMnO_4 began. One “quick-and-dirty” titration was performed to save time and to approximate the volume of potassium permanganate needed to react completely with the iron oxalate solution. The titration endpoint was reached when the solution turned and stayed pink for at least 30 seconds. The end volume of KMnO_4 was recorded, and the approximate volume of KMnO_4 used to complete the titration was calculated. The titration steps were then repeated for each of the remaining three flasks. The leftover waste solutions were then neutralized and disposed of down the drain. Any permanganate remaining in the buret was neutralized and combined it with the oxalate waste in the waste bottle in the fume hood.

3D. Quantitative Analysis of the Iron (III) Ion through Solution Spectrophotometry

0.1200g of $[\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}]$ was dissolved in the corresponding volume of deionized water in a 100 mL volumetric flask, giving a solution of approximately 0.003M in iron (II). In a 50 mL volumetric flask, 50 mg of $\text{C}_6\text{H}_8\text{O}_6$ and 15 mL of deionized water were combined, and the solution was swirled to dissolve the solid. Using a 2 mL pipet, 2 mL of the standard iron solution was transferred from the 100 mL flask into the 50 mL flask, along with 15 mL of the 0.0020M bipyridyl solution. The solution was then thoroughly mixed by inverting the sealed flask several times. Finally, the dilution was completed by filling the 50 mL volumetric flask to its calibration line with pH 4.7 buffer. This solution (A) was the “known” which would be used to determine the iron concentration of the complex. The solution was then transferred to a clean and dry 125 mL Erlenmeyer flask and sealed.

0.1050g of the synthesized green crystals were measured and combined with 20 mL of deionized water and 5 mL of 9M H_2SO_4 in a 50 mL disposable centrifuge tube, swirling to allow the crystals to dissolve. 15 mL of 0.5M CaCl_2 were then added and the solution was allowed stand for at least 5 minutes for complete precipitation. The tube was then centrifuged for 2 minutes in a balanced centrifuge. Using a dropper, all the supernatant liquid was transferred to a 100 mL volumetric flask. The solution was then

diluted to the calibration line with deionized water and mixing thoroughly. The centrifuge tube was then recapped and disposed of. 80 mg of solid ascorbic acid, 2 mL of the solution from the 100 mL flask (of supernatant & water), 20 mL of the 0.0020M bipyridyl solution, and 20 mL of the pH 4.7 buffer were then added to a 50 mL volumetric flask. The final solution pH was checked to be approximately 5.0, and more buffer was added if necessary. The solution was then diluted to the calibration line with deionized water and thoroughly mixed. This solution (B) was the “unknown” which would be used to determine the iron concentration of the complex. This solution was then transferred to a clean and dry 125 mL Erlenmeyer flask and sealed.

Using the template shown below, two sets of dilution samples were prepared – one with the known solution (A) and one with the unknown (B), resulting in a total of ten samples. Ten small clean and dry test tubes were marked with the appropriate labels.

<i>Dilution Sample</i>	<i>Volume of Fe^{2+} Solution</i>	<i>Volume of H_2O</i>	<i>Total Volume</i>
A1 & B1	0.75 mL	6.75 mL	7.50 mL
A2 & B2	1.50 mL	6.00 mL	7.50 mL
A3 & B3	2.25 mL	5.25 mL	7.50 mL
A4 & B4	3.75 mL	3.75 mL	7.50 mL
A5 & B5	5.25 mL	2.25 mL	7.50 mL

To avoid evaporation, test tubes were covered until ready for use. The Spectronic 20 was turned on, the wavelength was set to 520 nm, and 15 minutes were given for warming up. 2 small test tubes filled to 75% capacity with deionized water were found and tested to ensure that both provided the same percent transmission in the spectrophotometer, ensuring their proper matching. The instrument was then confirmed, and adjusted if necessary, to read 0%T (transmittance) when empty and 100%T when the test tube of water had been inserted. It may have been necessary to repeat the calibration several times to ensure the accuracy of the two test tubes, and later in the experiment as needed. Using a permanent pen, the orientations were marked so that the identical test tube setups could be made in future measurements. One of the test tubes was emptied, filled with the most dilute known solution, and inserted into the spectrophotometer, making sure to orient it correctly, and the %T was recorded. In between measurements, the accuracy of the spectrophotometer was checked with the un-emptied test tube of water – it should still have read 100%T. The procedure was then repeated for the remaining four test tubes of known solution, proceeding from the most dilute to the most concentrated. Then, after rinsing and drying the experimental test tube, the same procedure was performed for the unknown solutions, once again going from the most dilute to the most concentrated solutions. This was done to reduce the effects of solution carryover. The %T's of the known and unknown samples were then compared – if there were fewer than 3 unknown values which lay within the range of the known %T's, new dilutions had to be prepared, and more measurements taken, using the same Spectronic 20 and glassware. The waste solutions were then collected, neutralized if necessary, and disposed of down the drain.

4. Results

4A. Synthesis of the Iron-Oxalate Coordination Complex

Observations

The iron (II) ammonium sulfate, of chemical equation $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2$, was a light blue-green crystalline substance. Exactly 1.2000g of the hydrated salt was measured out and combined with 3.0 mL of deionized H_2O and 3 drops of 6M H_2SO_4 , a clear and colorless solution, in a 50 mL beaker. The solution was then immersed in a water bath and achieved a maximum heating temperature of 56.6°C, then was removed from the heat.

5.98 mL of 1M oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$, a clear and colorless solution) was then introduced into the same beaker, immediately producing a yellow precipitate of $\text{Fe}(\text{C}_2\text{O}_4)$ suspended – and later settling – in a clear solution with a goldenrod hue.

Upon testing the solution with pH paper, it was confirmed that the beaker contained “strongly acidic” solution. The beaker was then lowered into the same water bath and was allowed to boil for four minutes at a maximum temperature of 100°C . The solution was then removed from heat and allowed to cool for nine minutes. At this time, the digested precipitate had settled to the bottom of the baker, with a light, clear yellow solution on the top. A watch glass was placed over the mouth of the baker to keep out any possible impurities and contaminants. The decanted supernatant was a clear green-yellow solution, and a soggy mustard-colored heap of solid was left behind in the original beaker. The precipitate was washed with ~3 mL samples of water. The decanted liquid ranged from clear yellow to clear and colorless.

The potassium oxalate solution was clear and colorless. After 3.0 mL of it was added to the beaker, the contents turned an orangish-yellow. The beaker was heated to 42.0°C , and upon removal from the heat, 10 mL of 6% hydrogen peroxide (H_2O_2 , a clear and colorless solution) was slowly added. Small bubbles appeared with almost every drop, and the beaker turned warm to the touch, indicating the taking place of an exothermic reaction. The beaker contents soon turned a rusty, murky brown. The mixture was then heated to 100.0°C without noticeable change in appearance, but turned a clear green-yellow color upon the addition of 2.0 mL of 1M $\text{H}_2\text{C}_2\text{O}_4$. 145 drops of 95% ethanol (EtOH) were added, and crystallization initiated. The beaker was wrapped in foil and left in the lab drawer for a week. Upon return, a layer of small granulated radioactive-green crystals had settled to the bottom of the baker, which was also filled with a clear neon-green solution. Vacuum-filtering the crystals produced a clear yellow-green solution, which turned cloudy after a few minutes. The crystals were left in the crucible for use in the next part of the experiment.

Data

Mass of $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2$: 1.2000g

Mass of final product, lime-green crystals: 1.1####g

(Additional data may be found in the *Experiment 4A Lab Report*²)

Calculations

Equations Used:

$$\% \text{ yield} = \frac{\text{actual moles of product obtained}}{\text{maximum theoretical yield, in moles}} \times 100\%$$

$$\text{Actual moles of product} = \frac{\text{mass of crystals}}{\text{molar mass of crystals}}$$

$$\text{Theoretical product yield} = \frac{\text{mass of limiting reactant}}{\text{molar mass of limiting reactant}}$$

Limiting reactant is $\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$, molar mass of 392.15 g/mol

Mass of synthesized crystals = mass of filled crucible (g) – mass of empty crucible (g)

$$= 14.0389\text{g} - 12.0209\text{g}$$

$$= 1.1####\text{g}$$

Part A: If the product is $\text{K}[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_2] \cdot 2\text{H}_2\text{O}$ – molar mass is 307.03 g/mol

General Equation: $2\text{Fe}(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O} + \text{K}_2\text{C}_2\text{O}_4 + 3\text{H}_2\text{C}_2\text{O}_4 \rightarrow 2\text{K}[\text{Fe}(\text{C}_2\text{O}_4)_2] \cdot 2\text{H}_2\text{O}$

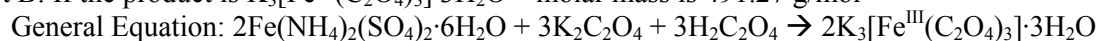
$$\text{Actual moles of product} = \frac{1.1180 \text{ g}}{307.03 \text{ g/mol}}$$

$$= 0.0036413 \text{ mol}$$

$$\begin{aligned} \text{Theoretical product yield} &= \frac{1.2000 \text{ g}}{392.15 \text{ g/mol}} \\ &= 0.0030601 \text{ moles beginning salt} \\ &= 0.0030601 \text{ moles final crystals (1:1 molar ratio, see equation)} \end{aligned}$$

$$\begin{aligned} \text{Percentage yield} &= \frac{0.0036413 \text{ mol}}{0.0030601 \text{ mol}} \times 100\% \\ &= 118.99\% \end{aligned}$$

Part B: If the product is $\text{K}_3[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ – molar mass is 491.27 g/mol



$$\begin{aligned} \text{Actual moles of product} &= \frac{1.1180 \text{ g}}{491.27 \text{ g/mol}} \\ &= 0.0022757 \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Theoretical product yield} &= \frac{1.2000 \text{ g}}{392.15 \text{ g/mol}} \\ &= 0.0030601 \text{ moles beginning salt} \\ &= 0.0030601 \text{ moles final crystals (1:1 molar ratio, see equation)} \end{aligned}$$

$$\begin{aligned} \text{Percentage yield} &= \frac{0.0022757 \text{ mol}}{0.0030601 \text{ mol}} \times 100\% \\ &= 74.367\% \end{aligned}$$

Since experimental values may not exceed 100% of the theoretical value, the 118.99% sample is rejected. Therefore, the product must be $\text{K}_3[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$.

4B. Analysis of the Oxalate Ligand Content through Redox Titration

Observations

KMnO_4 observations: opaque purple solution, similar to grape juice

After combining titrated solutions, waste turned a heme red

Data

Concentration of KMnO_4 : 0.0100M

Table of the Four Redox Titration Trials (Trial “A” being “quick-and-dirty”)

Action/ Measurement	Flask A	Flask B	Flask C	Flask D
mass crystals	0.1054 g	0.1052 g	0.1021 g	0.1017 g
vol of H_2O	30 mL	30 mL	30 mL	30 mL
vol of 6M H_2SO_4	5 mL	5 mL	5 mL	5 mL
analyte temp.	65.0°C	61.5°C	62.7°C	63.1°C
initial	0.35 mL	0.50 mL	0.00 mL	0.05 mL

<i>KMnO₄ vol.</i>				
<i>final KMnO₄ vol.</i>	26.22 mL	26.05 mL	25.11 mL	24.95 mL
<i>volume KMnO₄ used</i>	25.87 mL	25.55 mL	25.11 mL	24.90 mL
<i>endpoint color</i>	dusty rose to light pink to clear	medium rose to pink to clear	medium rose to pink to clear	medium rose to pink to clear

Calculations

Equations Used:

$$\text{mmol of oxalate in complex sample} = \frac{.01 \text{ mmol KMnO}_4}{1 \text{ mL}} \cdot \frac{\text{mL of KMnO}_4}{2 \text{ mmol KMnO}_4} \cdot \frac{5 \text{ mmol H}_2\text{C}_2\text{O}_4}{2 \text{ mmol KMnO}_4}$$

$$\text{wt \% of oxalate} = \frac{\text{mmol C}_2\text{O}_4}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol}} \cdot \frac{88.018 \text{ g C}_2\text{O}_4}{1 \text{ mol}} \cdot \frac{1}{\text{g of sample}}$$

$$\text{average} = \frac{\text{sum of all samples}}{\text{number of samples}}$$

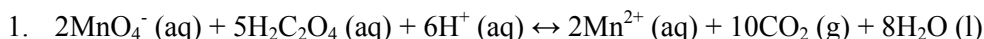
$$\text{standard deviation } (\sigma) = \sqrt{\frac{\sum (\bar{x} - x_i)^2}{N - 1}}$$

$$\text{mmol of oxalate in 100g complex} = \frac{100 \text{ g}}{\text{g of sample complex}} \cdot \frac{\text{mmol oxalate in sample}}{1 \text{ g of sample complex}}$$

Limiting reactant is Fe(NH₄)₂(SO₄)₂·6H₂O, molar mass of 392.15 g/mol

There is a 1:1 ratio between oxalic acid (H₂C₂O₄) and the oxalate ion (C₂O₄²⁻)

In calculations, H₂C₂O₄ is H₂C₂O₄; KMnO₄ is KMnO₄; C₂O₄ is C₂O₄²⁻;



2. Concentration of potassium permanganate (KMnO₄) solution: 0.0100M

3. Trial A

$$\frac{.0100 \text{ mmol KMnO}_4}{1 \text{ mL}} \cdot \frac{25.87 \text{ mL KMnO}_4}{2 \text{ mmol KMnO}_4} \cdot \frac{5 \text{ mmol H}_2\text{C}_2\text{O}_4}{2 \text{ mmol KMnO}_4} = .647 \text{ mmol C}_2\text{O}_4$$

Trial B

$$\frac{.0100 \text{ mmol KMnO}_4}{1 \text{ mL}} \cdot \frac{25.55 \text{ mL KMnO}_4}{2 \text{ mmol KMnO}_4} \cdot \frac{5 \text{ mmol H}_2\text{C}_2\text{O}_4}{2 \text{ mmol KMnO}_4} = .639 \text{ mmol C}_2\text{O}_4$$

Trial C

$$\frac{.0100 \text{ mmol KMnO}_4}{1 \text{ mL}} \cdot \frac{25.11 \text{ mL KMnO}_4}{2 \text{ mmol KMnO}_4} \cdot \frac{5 \text{ mmol H}_2\text{C}_2\text{O}_4}{2 \text{ mmol KMnO}_4} = .628 \text{ mmol C}_2\text{O}_4$$

Trial D

$$\frac{.0100 \text{ mmol KMnO}_4}{1 \text{ mL}} \cdot \frac{24.90 \text{ mL KMnO}_4}{2 \text{ mmol KMnO}_4} \cdot \frac{5 \text{ mmol H}_2\text{C}_2\text{O}_4}{2 \text{ mmol KMnO}_4} = .623 \text{ mmol C}_2\text{O}_4$$

4. Trial A

$$\frac{.647 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol}} \cdot \frac{88.018 \text{ g C}_2\text{O}_4}{1 \text{ mol}} \cdot \frac{1}{.1054 \text{ g sample}} \times 100 = 54.03\% \text{ C}_2\text{O}_4$$

Trial B

$$\frac{.639 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol}} \cdot \frac{88.018 \text{ g C}_2\text{O}_4}{1 \text{ mol}} \cdot \frac{1}{.1052 \text{ g sample}} \times 100 = 53.46\% \text{ C}_2\text{O}_4$$

Trial C

$$\frac{.628 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol}} \cdot \frac{88.018 \text{ g C}_2\text{O}_4}{1 \text{ mol}} \cdot \frac{1}{.1021 \text{ g sample}} \times 100 = 54.14\% \text{ C}_2\text{O}_4$$

Trial D

$$\frac{.623 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol}} \cdot \frac{88.018 \text{ g C}_2\text{O}_4}{1 \text{ mol}} \cdot \frac{1}{.1017 \text{ g sample}} \times 100 = 53.92\% \text{ C}_2\text{O}_4$$

Average Weight Percentage of Oxalate Ion

$$\frac{54.03 + 53.46 + 54.14 + 53.92}{4} = 53.89\%$$

Standard Deviation

$$\sigma = \sqrt{\frac{(53.89 - 54.03)^2 + (53.89 - 53.46)^2 + (53.89 - 54.14)^2 + (53.89 - 53.92)^2}{4 - 1}}$$

$$\sigma = .2988$$

$$2\sigma = .5976$$

Average Weight Percentage $\pm 2\sigma$:

$$53.89 \pm .5976\%$$

5. Trial A

$$\frac{100.0 \text{ g}}{.1054 \text{ g complex}} \cdot \frac{.647 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} = 614 \text{ mmol C}_2\text{O}_4 \text{ per } 100 \text{ g complex}$$

Trial B

$$\frac{100.0 \text{ g}}{.1052 \text{ g complex}} \cdot \frac{.639 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} = 607 \text{ mmol C}_2\text{O}_4 \text{ per } 100 \text{ g complex}$$

Trial C

$$\frac{100.0 \text{ g}}{.1021 \text{ g complex}} \cdot \frac{.628 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} = 615 \text{ mmol C}_2\text{O}_4 \text{ per } 100 \text{ g complex}$$

Trial D

$$\frac{100.0 \text{ g}}{.1017 \text{ g complex}} \cdot \frac{.623 \text{ mmol C}_2\text{O}_4}{1000 \text{ mmol}} = 613 \text{ mmol C}_2\text{O}_4 \text{ per } 100 \text{ g complex}$$

Average Number of mmol of Oxalate in 100g of Complex

$$\frac{614 + 607 + 615 + 613}{4} = 612 \text{ mmol C}_2\text{O}_4$$

Standard Deviation

$$\sigma = \sqrt{\frac{(612 - 614)^2 + (612 - 607)^2 + (612 - 615)^2 + (612 - 613)^2}{4 - 1}}$$

$$\sigma = 3.59$$

$$2\sigma = 7.18$$

Average Number of mmol of Oxalate $\pm 2\sigma$:

$$612 \pm 7.18 \text{ mmol C}_2\text{O}_4^{2-}$$

4C. Quantitative Analysis of the Iron (II) Ion through Solution Spectrophotometry

Observations

Solution turned red after addition of 15 mL of .0020M bipyridyl solution

Part II supernatant: clear yellow solution

Data

Part I: Preparation of a Standard Solution of Iron (II) Bipyridyl Complex

Mass of solid $\text{Fe}[(\text{NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}]$: 0.1205 g

Mass of ascorbic acid: 0.0532 g

Part II: Preparation of the Iron (III) Oxalate Complex for Measurement

Mass of green synthesized crystals: 0.1000 g (exactly!)

Mass of ascorbic acid: 0.0845 g

Part III: Spectrophotometric Measurements

Wavelength set to 520 nm

Fe^{2+} and Bipyridyl Dilution – Known Solution

Tube	Fe^{2+} volume	H_2O volume	Total volume	%T	Absorbance
A1	.75	6.75	7.50	72%	.1427
A2	1.50	6.00	7.50	61%	.2147
A3	2.25	5.25	7.50	43%	.3665
A4	3.75	3.75	7.50	18%	.7447
A5	5.25	2.25	7.50	5%	1.3010

Fe^{3+} Oxalate Dilution – Unknown Solution

Tube	Fe^{2+} volume	H_2O volume	Total volume	%T	Absorbance
B1	.75	6.75	7.50	84%	.0757
B2	1.50	6.00	7.50	75%	.1249
B3	2.25	5.25	7.50	60%	.2218
B4	3.75	3.75	7.50	42%	.3768
B5	5.25	2.25	7.50	25%	.6020

Calculations

Equations Used

$$A = \log\left(\frac{100}{\%T}\right), \text{ where } A = \text{absorbance and } T = \% \text{ transmittance}$$

$$\text{concentration} = \frac{\text{absorbance} + .1528}{14519}$$

$$\text{average} = \frac{\text{sum of all samples}}{\text{number of samples}}$$

$$\text{standard deviation } (\sigma) = \sqrt{\frac{\sum (\bar{x} - x_i)^2}{N - 1}}$$

$$1. \frac{.1205 \text{ g solid iron (II) ammonium sulfate}}{392.2 \text{ g}} \cdot \frac{1 \text{ mol}}{.100 \text{ L}} = 3.072 \times 10^{-3} \text{ M in iron (II)}$$

$$2. \frac{3.072 \times 10^{-3} \text{ mol}}{1 \text{ L}} \cdot \frac{.002 \text{ L}}{.05 \text{ L}} = 1.229 \times 10^{-4} M \text{ of } \text{Fe}^{2+} / \text{bipy}$$

3. Known Sample

Sample	M of Fe^{2+} /bipy		Dilution Conc.		Final Concentration
A1	1.229×10^{-4}	x	(.75mL/7.50mL)	=	$1.229 \times 10^{-5} M$
A2	1.229×10^{-4}	x	(1.50mL/7.50mL)	=	$2.458 \times 10^{-5} M$
A3	1.229×10^{-4}	x	(2.25mL/7.50mL)	=	$3.687 \times 10^{-5} M$
A4	1.229×10^{-4}	x	(3.75mL/7.50mL)	=	$6.145 \times 10^{-5} M$
A5	1.229×10^{-4}	x	(5.25mL/7.50mL)	=	$8.603 \times 10^{-5} M$

4. Fe^{2+} Bipyrimidyl Dilution (Known)

Tube	%T	Absorbance
A1	72%	.1427
A2	61%	.2147
A3	43%	.3665
A4	18%	.7447
A5	5%	1.3010

Fe^{3+} Oxalate Dilution (Unknown)

Tube	%T	Absorbance
B1	84%	.0757
B2	75%	.1249
B3	60%	.2218
B4	42%	.3768
B5	25%	.6020

For the "Beer's Law Calibration Graph," please refer to the attached page...

Also, note that the graph excludes trials B1 & B5 because their values are outside the desired range of .2-.8 absorbance values...

5. The equation of concern (obtained from trend line of previous graph): $y = 14519x - 0.1528$, where "y" represents the absorbance and "x" the concentration in mol/L.

Sample	Equation with Unknown A	Solve for x...	Final Concentration (M)
B1	$0.0757 = 14519x - 0.1528$	...	$x = 1.574 \times 10^{-5} M$
B2	$0.1249 = 14519x - 0.1528$	...	$x = 1.913 \times 10^{-5} M$
B3	$0.2218 = 14519x - 0.1528$	...	$x = 2.580 \times 10^{-5} M$
B4	$0.3768 = 14519x - 0.1528$	...	$x = 3.648 \times 10^{-5} M$
B5	$0.6020 = 14519x - 0.1528$	...	$x = 5.199 \times 10^{-5} M$

6. Concentration of Fe^{3+} in the 100 mL volumetric flask

Sample B1

$$\frac{1.574 \times 10^{-5} M}{1 \text{ mL}} \cdot \frac{10 \text{ mL}}{2.0 \text{ mL}} = 3.935 \times 10^{-5} M$$

Sample B2

$$\frac{1.913 \times 10^{-5} M}{2 \text{ mL}} \cdot \frac{10 \text{ mL}}{2.0 \text{ mL}} \cdot \frac{50 \text{ mL}}{2.0 \text{ mL}} = 2.391 \times 10^{-3} M$$

Sample B3

$$\frac{2.580 \times 10^{-5} M}{3 \text{ mL}} \cdot \frac{10 \text{ mL}}{2.0 \text{ mL}} \cdot \frac{50 \text{ mL}}{2.0 \text{ mL}} = 2.150 \times 10^{-3} M$$

Sample B4

$$\frac{3.648 \times 10^{-5} M}{4 \text{ mL}} \cdot \frac{10 \text{ mL}}{2.0 \text{ mL}} \cdot \frac{50 \text{ mL}}{2.0 \text{ mL}} = 2.280 \times 10^{-3} M$$

Sample B5

$$\frac{5.199 \times 10^{-5} M}{5 \text{ mL}} \cdot \frac{10 \text{ mL}}{2.0 \text{ mL}} \cdot \frac{50 \text{ mL}}{2.0 \text{ mL}} = 2.600 \times 10^{-3} M$$

7. (Unknown samples B1 and B2 were omitted because their values were not within the desired range of 0.2-0.8)

$$\bar{x} = \frac{.002150 + .002280 + .002600}{3} = .002343 M$$

8. mmol of Fe^{3+} (averaged with 2 standard deviations) in the ~0.1g sample, excluded B1 & B2 for reasons above...

Sample B3

$$(.002150 \text{ mmol}) \times (100 \text{ mL}) = .2150 \text{ mmol Fe}^{3+}$$

Sample B4

$$(.002280 \text{ mmol}) \times (100 \text{ mL}) = .2280 \text{ mmol Fe}^{3+}$$

Sample B5

$$(.002600 \text{ mmol}) \times (100 \text{ mL}) = .2600 \text{ mmol Fe}^{3+}$$

Average Number Of mmol Of Fe^{3+} In The 100 mL Of Solution

$$\bar{x} = \frac{.2150 + .2280 + .2600}{3} = .2343 \text{ mmol}$$

Standard Deviation

$$\sigma = \sqrt{\frac{(.2343 - .2150)^2 + (.2343 - .2280)^2 + (.2343 - .2600)^2}{3 - 1}}$$

$$\sigma = .0232$$

$$2\sigma = .0464$$

Average Number of mmol of $\text{Fe}^{3+} \pm 2\sigma$:

$$.2343 \pm .0464 \text{ mmol Fe}^{3+}$$

9. Calculating average percent (wt%) composition ($\pm 2\sigma$) of iron in original sample...

Sample B3

$$\frac{.2150 \text{ mmol Fe}^{3+}}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol Fe}} \cdot \frac{55.85 \text{ g Fe}}{1 \text{ mol Fe}} \cdot \frac{1}{.1000 \text{ g sample}} \times 100 = 12.01\%$$

Sample B4

$$\frac{.2280 \text{ mmol Fe}^{3+}}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol Fe}} \cdot \frac{55.85 \text{ g Fe}}{1 \text{ mol Fe}} \cdot \frac{1}{.1000 \text{ g sample}} \times 100 = 12.73\%$$

Sample B5

$$\frac{.2600 \text{ mmol Fe}^{3+}}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol Fe}} \cdot \frac{55.85 \text{ g Fe}}{1 \text{ mol Fe}} \cdot \frac{1}{.1000 \text{ g sample}} \times 100 = 14.52\%$$

Average Percentage Weight

$$\bar{x} = \frac{12.01 + 12.73 + 14.52}{3} = 13.09\%$$

Standard Deviation

$$\sigma = \sqrt{\frac{(13.09 - 12.01)^2 + (13.09 - 12.73)^2 + (13.09 - 14.52)^2}{3 - 1}}$$

$$\sigma = 1.29$$

$$2\sigma = 2.58$$

Average Weight Percentage of $\text{Fe}^{3+} \pm 2\sigma$:

$$13.09 \pm 2.58 \% \text{ Fe}^{3+}$$

10. Millimoles of Fe^{3+} Present in 100g of Complex

Sample B3

$$\frac{12.01\%}{100} \cdot \frac{100 \text{ g}}{1} \cdot \frac{1 \text{ mol Fe}}{55.85 \text{ g}} \cdot \frac{1000 \text{ mmol}}{1 \text{ mol}} = 215.040 \text{ mmol Fe(III) in 100g complex}$$

Sample B4

$$\frac{12.73\%}{100} \cdot \frac{100 \text{ g}}{1} \cdot \frac{1 \text{ mol Fe}}{55.85 \text{ g}} \cdot \frac{1000 \text{ mmol}}{1 \text{ mol}} = 227.932 \text{ mmol Fe(III) in 100g complex}$$

Sample B5

$$\frac{14.52\%}{100} \cdot \frac{100 \text{ g}}{1} \cdot \frac{1 \text{ mol Fe}}{55.85 \text{ g}} \cdot \frac{1000 \text{ mmol}}{1 \text{ mol}} = 259.982 \text{ mmol Fe(III) in 100g complex}$$

Average mmol of Fe^{3+} in 100g of Complex

$$\bar{x} = \frac{215.040 + 227.932 + 259.982}{3} = 234.318 \text{ mmol}$$

Standard Deviation

$$\sigma = \sqrt{\frac{(234.318 - 215.040)^2 + (234.318 - 227.932)^2 + (234.318 - 259.982)^2}{3 - 1}}$$

$$\sigma = 23.14$$

$$2\sigma = 46.28$$

Average mmol of Fe^{3+} in 100g of Complex $\pm 2\sigma$:

$$234.318 \pm 46.28 \text{ mmol Fe}^{3+}$$

4D. Wrap-Up Calculations for Complex: Experiments 4A-C

- A. 61# mmol $\text{C}_2\text{O}_4^{2-}$ roughly a 1:3 ratio between $\text{Fe}^{3+}:\text{C}_2\text{O}_4^{2-}$
 23# mmol Fe^{3+}
 $x=1$ and $y=3$
- B. $\text{K}_w[\text{Fe}_x(\text{C}_2\text{O}_4)_y] \cdot z\text{H}_2\text{O}$
 1 Fe^{3+} gives +3...3 $\text{C}_2\text{O}_4^{2-}$ give -6... H_2O is neutral...overall charge of -3
 Since K^+ provides charge of +1, need 3 potassium ions
 $W=3$

C. $K_3[Fe(C_2O_4)_3]$

$$\frac{234.318 \text{ mmol Fe}}{1 \text{ mol Fe}} \cdot \frac{3 \text{ mol K}}{1 \text{ mol Fe}} = 702.954 \text{ mol K} + \text{in } 100\text{g}$$

D. $\frac{234.318 \text{ mmol Fe}}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol Fe(III)}} \cdot \frac{55.85\text{g}}{1 \text{ mol Fe(III)}} = 13.09\text{g Fe(III)}$

$$\frac{702.954 \text{ mmol K}}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol K}} \cdot \frac{39.10\text{g}}{1 \text{ mol K}} = 27.49\text{g K} +$$

$$\frac{612 \text{ mmol oxalate}}{1000 \text{ mmol}} \cdot \frac{1 \text{ mol}}{1 \text{ mol Fe(III)}} \cdot \frac{88.018\text{g}}{1 \text{ mol Fe(III)}} = 53.87\text{g oxalate}$$

$$13.09 + 27.49 + 53.87 = 94.45$$

$$100 - 94.45 = 5.55$$

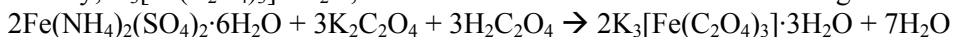
$$\frac{5.55\text{g H}_2\text{O}}{18.02\text{g}} \cdot \frac{1 \text{ mol H}_2\text{O}}{1 \text{ mol}} \cdot \frac{1000 \text{ mmol}}{1 \text{ mol}} = 307.99 \text{ mmol H}_2\text{O} \dots 2:1 \text{ ratio in H}_2\text{O:Fe}^{3+}$$

$z = 2$

E. Determined Empirical Formula:

$K_3[Fe(C_2O_4)_3] \cdot 2H_2O$...does not correspond to the real empirical formula

F. Since E's empirical formula does not correspond to postlab 2g's empirical formula, the molecular formula and molecular mass to be used in this calculation should reflect the "proper" product – namely, $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$, which has a molecular mass of 491.249 g/mol.



$$\frac{1.2000\text{g Fe(NH}_4)_2(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}}{392.15\text{g}} \cdot \frac{1 \text{ mol}}{2 \text{ mol iron salt}} \cdot \frac{2 \text{ mol potassium salt}}{1 \text{ mol}} \cdot \frac{491.249\text{g}}{1 \text{ mol}} = 1.503\text{g prod}$$

$$\text{Percentage Yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$= \frac{1.1180\text{g}}{1.503\text{g}} \times 100 = 74.38\%$$

5. Discussion

4A – Synthesis of an Iron (III) Oxalate Complex

In the synthesis of the iron (III) oxalate complex, 1.1###g of the final lime-green crystals were obtained. The percent yield of $K[Fe^{3+}(C_2O_4)_2] \cdot 2H_2O$ was calculated to be 11#.#%, and the percent yield of $K_3[Fe^{3+}(C_2O_4)_3] \cdot 3H_2O$ was calculated to be 74.###%. Since the experimental values may not exceed 100% of the calculated theoretical value (which assumes ideal conditions), the possibility of the complex being $K[Fe^{3+}(C_2O_4)_2] \cdot 2H_2O$ is thrown out, leaving $K_3[Fe^{3+}(C_2O_4)_3] \cdot 3H_2O$ as the correct one. There are no averaged values because only one trial was performed.

Apart from the usual human-mediated errors in the measuring and distribution of chemicals and in the readings of instrumental information – both of which are of much concern in this particular experiment, given that the experimental procedure contained little else but sequential measurements and readings – another source of error involved the maintenance of the solution pH level. In step two, the pH was checked to be strongly acidic. Such an environment would also result in an increased presence of the oxalate ion (see equation two, from Introduction), which would then be available for coordination with the Fe^{3+} ion. Had the environment been insufficiently acidic, equation two's equilibrium would have favoured the reactant side, resulting in an excess of oxalic acid ($H_2C_2O_4$), which is useless for complex ion formation. Equation two's equilibrium is carefully maintained by balancing the stoichiometric

amount of oxalic acid and aqueous potassium oxalate ($\text{K}_2\text{C}_2\text{O}_4$, which dissolves in solution to give the oxalate ion, $\text{C}_2\text{O}_4^{2-}$).

Additional sources of error involved dirty and/or improperly calibrated glassware, failure to completely encase the crystals in foil (they're light-sensitive), improper cleaning of the crucible (which would lead to inaccurate filtration), contaminated or already-catalyzed H_2O_2 (which would result in the hindered oxidation of Fe^{2+} to Fe^{3+}), and failure to allow the crystals to completely air-dry (mass of H_2O would have accidentally been included, resulting in an overestimation of the percentage yield).

Although the calculated percentage yield of $\text{K}_3[\text{Fe}^{3+}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ was sufficiently high enough to conclude that this experimental procedure is both an effective and successful way to synthesize, isolate, and purify an iron (III) oxalate compound, it was noted that the filtrant liquid in step 6 eventually turned cloudy and appeared as if there was a minimal sprinkling of crystals in the bottom of the filter flask. These crystals may have been lost in the filtration process by the use of the 95% EtOH. Had the EtOH been more highly concentrated, the crystals would probably have remained in the crucible, seeing as they were water-soluble and may have dissolved in the 5% of H_2O present in the ethanol mixture.

4B – Analysis of the Oxalate Ligands in the Iron (III) Oxalate Complex

In the analysis of the green synthesis product for oxalate content, it was found that the four trials yielded calculations of 61#, 60#, 61#, and 61# mmol of $\text{C}_2\text{O}_4^{2-}$ per 100g of complex analyzed. The averaged value, with the inclusion of two standard deviations, reported to be $61\# \pm 7.##$ mmol of $\text{C}_2\text{O}_4^{2-}$ per 100g of complex analyzed. The relative small size of the standard deviations (1.17% of the average) suggest that the titrations were performed very precisely with one another, and such a conclusion is consistent with the experimental findings as well. The millimoles of oxalate within the crystals were .64#, .63#, .62#, and .62#, for trials A-D, respectively. The weight percentages for each trial (A-D) were 54.0#%, 53.4#%, 54.1#%, and 53.##%, with an average of $53.## \pm .5###\%$. The standard deviation in this case was 1.11% of the average value, also indicating that the samples are very precise in measurement. No obvious outliers were identified, so the Q-Test was skipped and no extra trials were attempted.

Possibility for error included the initial mass-recordings of the crystals; even small amounts of crystals may have adhered to the weighing paper, and in the transfer of the crystals from the balance to the flask, air currents and unsteady movements may have also caused the loss of material. A possible remedy for this would be to “mass by difference,” the technique of which was practiced in an earlier laboratory experiment.

Also, realizing that the buret is a rather large and unwieldy instrument, it may have been very possible for some of the permanganate analyte to have been lost in the filling and utilizing of the buret. A leaking stopcock would also be a source of error. A very common source of error with burets and titration involves overshooting the endpoint – when the endpoint is fast approaching, even the extra addition of a drop or two could turn the solution from clear into a permanent pink. The “quick and dirty” titration was designed with student impatience in mind; after one “q&d” trial was performed, an approximation for the volume of KMnO_4 needed could be made, and the rest of the titrations could proceed more smoothly.

Minor sources of error may have involved: impurity of crystals, improper massing of the crystals, dirty and/or contaminated glassware, improperly calibrated instruments, failing to record the initial volume of the titrant, inconsistent readings of the buret (i.e., above eye level, below, etc), the interaction of the CO_2 of the atmosphere on the relative acidity of the analyte (resulting in more KMnO_4 used than was actually needed), and general calculation errors.

Based upon the high precision of the results, it is feasible to suggest that redox titrations are both efficient and accurate techniques by which to determine the content of a particular soluble species (here: oxalate ion) in a solution. The method is not as time consuming as gravimetric analysis but is much more practical, allowing multiple tests to be run in a short amount of time – incentives for any chemist as well as student.

4C – Analysis of the Iron in the Iron (III) Oxalate Complex

The concentrations of Fe^{3+} within the 100 mL volumetric flasks were calculated to be, in order of procedure, $3.935 \times 10^{-3} M$, $2.391 \times 10^{-3} M$, $2.150 \times 10^{-3} M$, $2.280 \times 10^{-3} M$, and $2.600 \times 10^{-3} M$, with an average of $0.002343 M$. The millimoles of Fe^{3+} calculated to be in the initial $\sim 0.1 \text{ g}$ sample were found to be, in unknown samples 3-5, $.2150 \text{ mmol Fe}^{3+}$, $.2280 \text{ mmol Fe}^{3+}$, and $.2600 \text{ mmol Fe}^{3+}$, with an average of $.2343 \pm .0464 \text{ mmol Fe}^{3+}$. The average weight percentages of Fe^{3+} for samples 3-5 of the unknowns were found to be 12.###%, 12.7#%, and 14.###%, with an average weight percentage of $13.09 \pm 2.58\%$. The number of millimoles present in 100g of complex were found to be, for unknown samples 3-5: $215.### \text{ mmol Fe}^{3+}$, $227.### \text{ mmol Fe}^{3+}$, and $259.### \text{ mmol Fe}^{3+}$, with an average of $23\#.### \pm 4\#.## \text{ mmol Fe}^{3+}$ in every 100g of complex.

Firstly, the Beer's Law graph could only reflect the innermost 3 values of the known dilution, for the first and last trials had absorbance values outside the "suggested" values of .2000 to .8000. However, a decent trend line was able to be drawn, and throughout the course of the calculations, the elimination of the two data points did not seem to have much effect on the outcome of the rest of the experiment. Likewise, two trials from the unknown had to be eliminated, for the first and second trials had absorbencies below the recommended range of .2000 to .8000.

There were many sources of error in this experiment. First of all, the dilutions themselves were prone to many mistakes, with the possibility of adding incorrect volumes to incorrect test tubes very great. Incorrectly reading and using the pipet could also be a source of error. If the crystals had not been sufficiently dried before use, then an excess of water would affect the final calculation of the amount of water within the complex itself. Incorrectly zeroing the Spectronic 20 would have resulted in the inconsistency of readings, as would the gradual dirtying of the test tubes. Even though gloves were worn, there were still plenty of opportunities for the test tubes to pick up additional dust and grime along the duration of the experiment, thereby affecting the absorbency readings.

I may have accidentally removed an excess of water from my complex without meaning to; I stored my crystals in my dessicator for the better part of an hour before I realized that the dessicant would be likely to have a negative effect on the water content of my hydrated iron oxalate salt. In a slightly less elaborate possibility, the beaker of crystals may not have been fully capped, and water may simply have evaporated out of the holding container.

The spectrophotometer readings were not very reliable, as the instruments seem to be a bit outdated and had a rather nasty habit of fluctuating in readings, until it was decided that a bit of accuracy could be sacrificed in order for everyone in the classroom to finish the procedure on time. Perhaps if there were more instruments available, or better attention and vigilance had been applied towards the preparation of the solutions, then Spectrophotometry would have resulted in more successful readings. The empirical formula was different from that of the "true" compound mentioned in the postlab; had there been a better alternate method to determining the iron (III) content within the complex, then the empirical formula would have probably been correctly identified and in case of an experiment concerning the identification of an unknown, the unknown would have been properly identified as well.

¹ Dr. Ian Ball, *Chemistry 6BL Laboratory Manual*, University of California, San Diego, ##### Quarter 200#

² My name, *Expt. 4A: Synthesis of Iron (III) Oxalate Complex*, November ##, 200#